Aanal Sonara

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Education

2025 - 2027* MS Robotics | **University of Pennsylvania** (GPA: 4.0/4.0)
Coursework - *Neural Representations & Neural Scene Rendering**, Learning in Robotics, F1/10 Autonomous Racing, Machine Perception, Introduction to Robotics, Computer Vision & Computational Photography

2019 - 2023 B.Tech. Electrical Engineering | **Indian Institute of Technology Bombay** (GPA: 3.26/4.0)
Minor in AI & Data Science
Coursework - Optimisation in ML, Advanced ML, Advanced Methods in Satellite Image Processing, Markov Chains & Queueing Systems, Automatic Speech Recognition, Natural Language Processing

Professional Experience

Autonomy Intern, Skydio HQ | San Mateo, USA May '26 - Aug '26
Engineered open-vocabulary object search and tracking algorithms running in real-time on cloud
- Implemented **SAM3 detector** & **tracker** modules & plugged **OWLv2** detector with SAM3 tracker for benchmark
- Redesigned SAM3 tracker's **state machine** to track on streaming video input by maintaining groups of detections
- Developed evaluation code to benchmark all variants of detector and tracker for specific prompt concepts
- Achieved F1-score **0.67**@mIoU **0.88** on 10 concept prompts on **640x360** images at **1.5** fps; deployed on cloud
- Spearheaded data collection of **24** flights on **3** sites & annotation of **10k+** object instances from **10** concepts

Engineer II, IdeaForge Technology | Bangalore, India Apr '24 - Aug '25
*Engineered and integrated perception algorithms on **Netra 5** and **Q6 V2** UAVs for **GNSS-denied** operation*
- Presented Ideaforge at **Indian Conference** on Computer Vision, Graphics & Image Processing (ICVGIP), 2024
- Deployed Day & Night Visual-SLAM on UAVs & achieved **RMSE**$\leq$ **5%** 100m altitude & range of upto 10kms
- Created Proof-of-concept for **Satellite Image** based Navigation that reduced the **relocalisation** error by **8%**
- Implemented dense **3D reconstruction** with **Gaussian-Splatting** optimised by novel frame rejection algorithm
- Improved V-SLAM **RMSE** by **3%** by creating map **saving** & **re-using** feature for upto 30kms flight path length

Systems Engineer, Enphase Energy | Bangalore, India Jul '23 - Mar '24
*Developed novel **Energy Management Systems** for Solar Energy Products to facilitate **reliable grid integration***
- Performed 20+ hardware & software tests on diverse system setups for **Predictive Analysis** of **10+** failure modes
- Monitored **50+** alpha and beta installations and validated the system in **real-world weather & grid conditions**

Research Intern, Sony HQ | Tokyo, Japan May '22 - Jun '22
*Devised an API to integrate **Relational Databases** with **Visual Navigation Algorithms** to handle large datasets*
- Successfully integrated **PostgreSQL** database with **Multi-Threaded** Visual Algorithms to eliminate **mutex errors**
- Achieved **8% computational efficiency boost** for **ORB-SLAM3** algorithm by deploying database integration API
- Reduced **Memory usage**, **CPU utilization**, **Latency**, & Translation Error by ~ 5% using **Binary String** datatype

Research & Technical Experience

Deep Event SLAM | Prof. Pratik Chaudhari, UPenn Jan '26 - May '26
- Adapted **3D Gaussian Splat**-based edge reconstruction model to **event** data with sparse 2D edge supervision
- Converted asynchronous events into edge images by adaptive exponential-decay filter & **PidiNet** edge detector
- Built a **parametric** edge extraction pipeline by clustering 3D Gaussian points & fitting Bezier curves per cluster
- Evaluated the pipeline on Habitat and UZH-FPV drone racing datasets comparing RGB and event input data

Deep Visual SLAM, UMIC | IIT Bombay Dec '21 - Apr '22
- Created a **ROS package** to generate 3D map and publish **real-time pose** for deep learning based Visual SLAM
- Trained the image feature description model **HFnet+NetVLAD** with Superpoint teacher model on **100K** images
- Fine-tuned Graph Neural Network model **SuperGlue** for feature matching with **70%** accuracy on training data
- Used **LinearPnP** on image & depth data from Zed2 camera to create sparse **3D map** and compute the drone pose

International Aerial Robotics Competition (AUVSI) | UMIC, IIT Bombay Aug '20 - Aug '22
- Trained YoLo_v4 model on $10k+$ simulation data & finetuned on $2k+$ real-world data to bridge sim2real gap
- Achieved **95%** mAP in real-time at 0.5 IoU with **20 fps** on YOLO_v4 deployed on **Nvidia Xavier NX** platform
- Optimized inference using **TensorRT** & **NVDLA** in **C++**; reduced CPU-GPU load, achieved **3×** faster inference
- Developed a visuomotor controller to align robotic manipulator with target object & perform replacement task

Unmanned Aerial Vehicle (UAV) Competition (ICUAS), UMIC | IIT Bombay Jan '23 - Jun '23

- Created **obstacle avoidance** module & **Waypoint** navigation trajectory to explore unknown indoor environment
- Fine-tuned **ORB-SLAM3** hyperparameters for localisation in semi-indoor environment using Intel RGBD sensor

Caption Generation From Images, B.Tech Project | Prof. B. Banerjee, IIT Bombay Aug '22 - Apr '23

- Fine-tuned CLIP on Satellite Image and caption data to minimise **Contrastive Loss** in **HuggingFace** framework
- Engineered a novel **hierarchical** feature extraction system using a **CLIP** image encoder and **GPT-2** text decoder
- Used **OpenAI** plugin for text **data augmentation** & image processing on images to increase dataset by **50%**
- Integrated CLIP's encoder with an **LSTM** model to enhance caption generation accuracy & **BLEU score** by **15%**

Academic Projects

F1/10th race overtaking using RL | Prof. Rahul Mangharam, UPenn Jan '26 - May '26

- Built 2D LiDAR-based local occupancy-grid & **Bresenham** line collision detector for reliable obstacle detection
- Designed hybrid controller to dynamically switch from pure pursuit to RL controller for overtaking obstacles
- Validated the RL overtaking controller in simulation and on physical F1/10th hardware across static/dynamic obstacles and left/right turns, achieving overtaking speeds up to **4.5** m/s (hardware) and **6** m/s (simulation)

Graspable regions from RGB images | Prof. Jianbo Shi, UPenn Aug '25 - Dec '25

- Used **SAM2** for segmenting the object of interest from image & **DepthAnythingV2** for calculating psuedo depth
- Created object mesh from InstantMesh & used Any6D model to get object pose & corrected mesh upto a scale
- Employed novel scoring system to create an affordance heatmap of mesh vertices & get stable graspable points
- Use the diagonally opposite point through center of mass for stable grip & fixed height box for graspable region

Cube Pick and Stack | Prof. Rachel Holladay, UPenn Nov '25 - Dec '25

- Explored motion planning algorithms like **RRT** and **Potential fields** in configuration space on Gazebo Simulation
- Implemented **Forward** and **Inverse** kinematics to pick static & dynamic cubes then stack; for Panda arm gripper
- Finetuned for hardware offset between camera and gripper; stacked **4** blocks in **5** minutes in hardware

Medical Image Captioning | Prof. Pushpak Bhattacharyya, IIT Bombay Jan '23 - Apr '23

- Built a medical image captioning model using a **Vision Transformer** (ViT-patch16-224) encoder & GPT2 decoder
- Improved **ROUGE score** by **15%** by paraphrasing using ChatGPT API & synonym exchange to augment data
- Achieved **20%** increase in medical report generation accuracy of finetuned ViT+GPT-2 architecture over LSTM

Script Generator, Introduction to Machine Learning Jan '21 - Apr '21

- Generated short script scenes using GPT2, trained on a lexically dense dataset containing **25,000+** unique words
- Observed **KL-divergence** of common words is higher in fine-tuned model of GPT-2 using HuggingFace Library

Ensemble Attack using Linear Backpropagation | Prof. Sunita Sarawagi, IIT Bombay Jan '22 - Apr '22

- Implemented **Linear BP** with **MI-FGSM** based adversarial attack for **black box** model with **ensemble** of models
- Improved success rate of attack upto **100%** in certain models as VGG19 and **transferability** of black box attack

Publications

Agarwal, M., **Sonara, A.**, Prakash, B., Khantwal, R., Patel, V., Naik, H., ... Shukla, D. A Novel Approach to Autonomous Aerial Manipulation using Multiple Drones. *2021*
12th Symposium on Dynamic Flight Behavior For Aerial Robotics *Link*

Achievements

- Ranked **1st in Simulation** Round & **3rd in Hardware** Round out of **39 global teams** at ICUAS, Warsaw *[2023]*
- Received **Design Award**, Scrutineering Award, and Advancement Award at the **UAS Challenge, UK** *[2023]*
- Declared **World Champions** at the **Internaional Aerial Robotics Competition** M9 Simulation Challenge *[2022]*
- Secured **World Rank 4** at the ASME Student Robotics Design Challenge among **100+** university teams *[2022]*

Teaching Experience

CIS5810 Computer Vision and Computational Photography Prof. Jianbo Shi, UPenn Aug '26 - Dec '26
MEAM5200 Introduction to Robotics Prof. Rachel Holladay, UPenn Aug '26 - Dec '26
CIS5800 Machine Perception Prof. Lingjie Liu, UPenn Jan '26 - May '26

Organisational Experience

Overall Coordinator UMIC, IIT Bombay May '22 - May '23

- Headed a **90 membered** funded student technical team developing state-of-the-art autonomous drones and cars
- Managed team operations, logistics, hardware & supervising a **budget** of INR **2M+** procured from IIT Bombay
- Represented team at exhibitions like Techfest & Tech Expo with a footfall of **10k+** students & leading industrials
- Diversified and widened the team network by connecting with over **100** global professors, other teams & alumni

Research Grants & Funding UMIC, IIT Bombay May '22 - May '23

- On-boarded **5+** Industrial Sponsors for hardware supply & technical assistance at Innovation Cell, IITB *[2023]*
- Secured INR **1.5M** funding from IIT Bombay for the **International Aerial Robotics Competition** (IARC) *[2021]*
- Raised a combined INR **2M** funding from IIT Bombay Alumni for Unmanned Aerial Systems Competition *[2023]*

Skills

Languages - Python, C++, SQL (RDBMS), MATLAB, VHDL, Embedded C
Libraries - TensorFlow, OpenCV, Pytorch, HuggingFace, JetPack, Keras, TensorRT, Docker, g2o, tracy
Micro-controllers - 8051, 8086, MIPS, Arduino UNO, ESP32
Edge Device - Nvidia Nano, Nvidia Xavier NX, RB5
Simulation Software - Gazebo, ROS (Robot Operating System), QGC (QGroundControl), Airsim